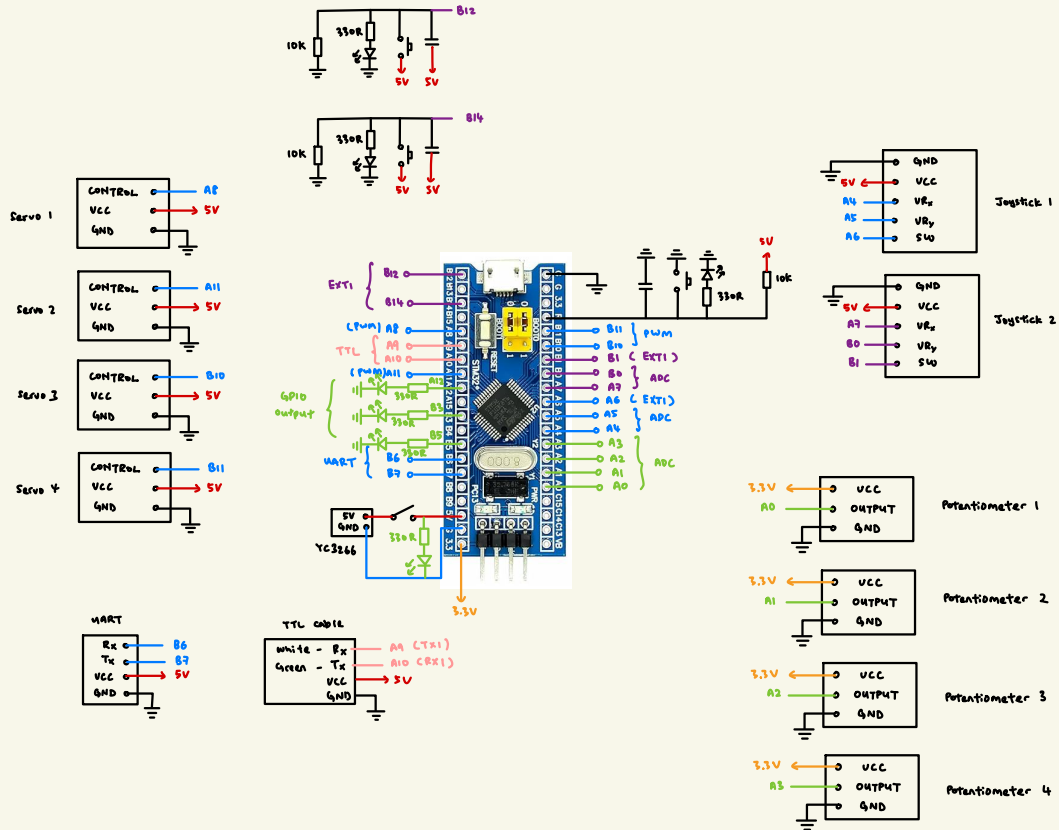
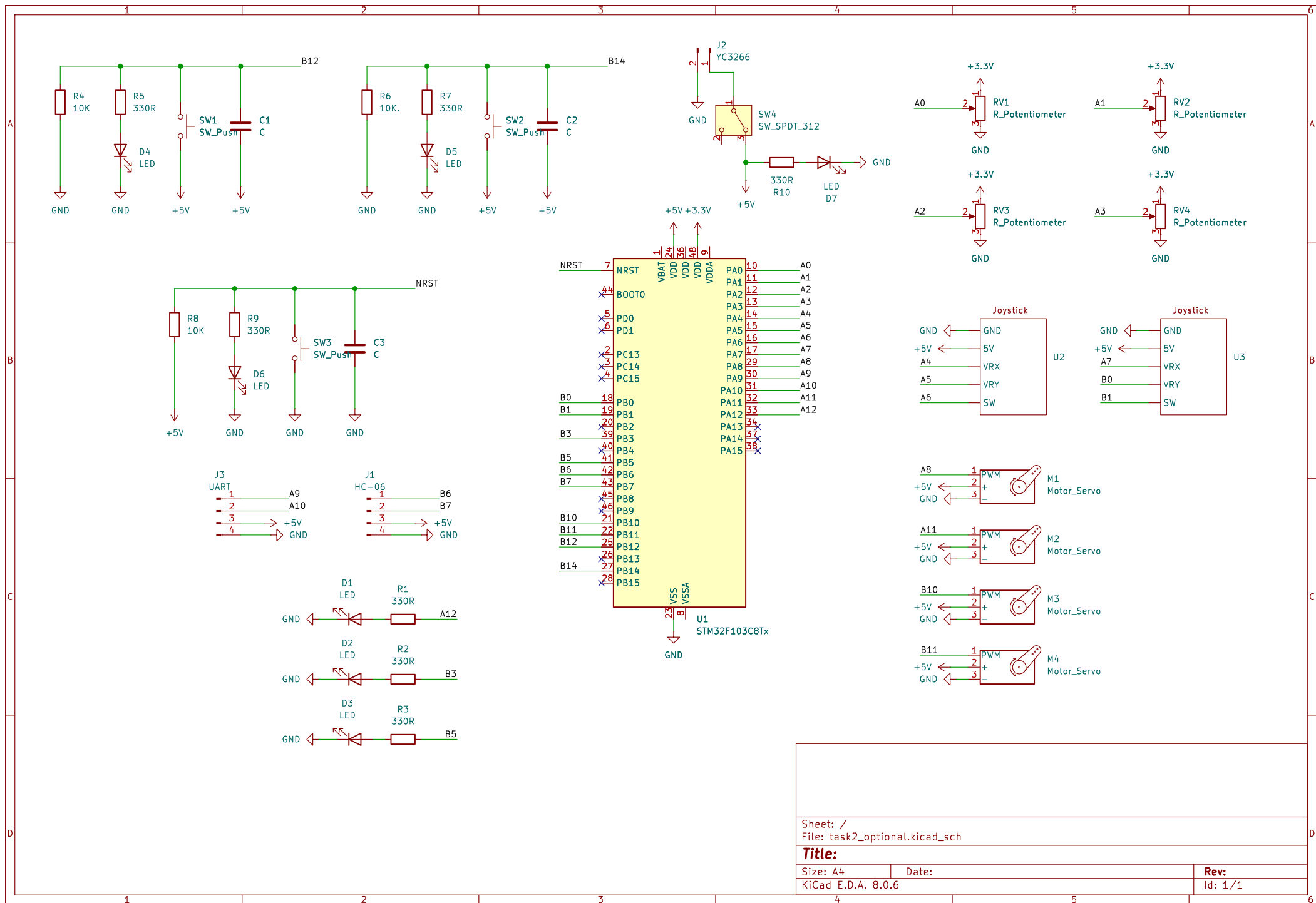






Sheet: /
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Title:

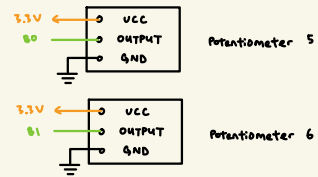
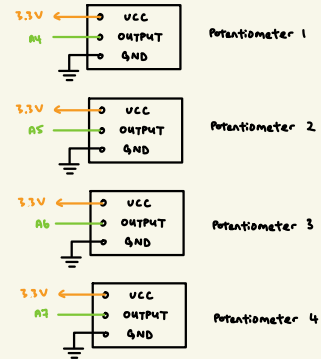
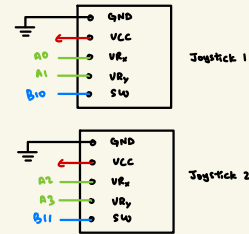
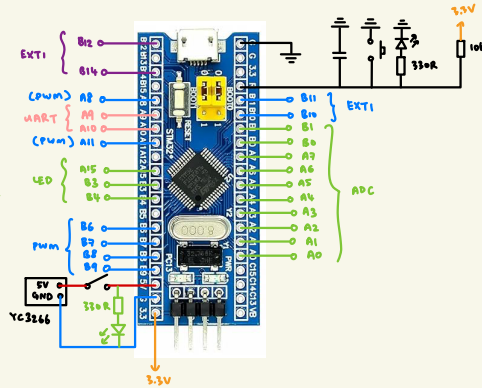
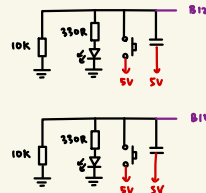
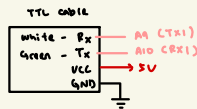
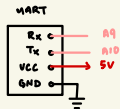
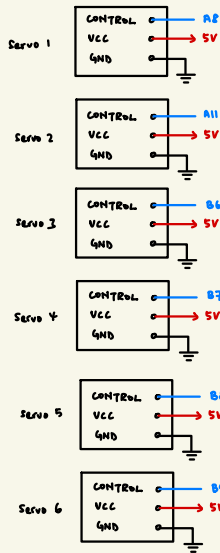
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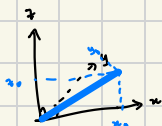
Date:

KiCad E.D.A. 8.0.6

Rev:

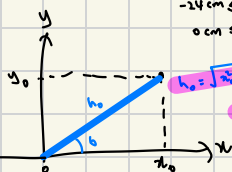
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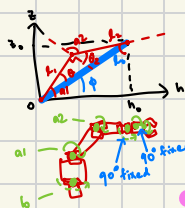
$$-24 \text{ cm} \leq x \leq 24 \text{ cm}$$

$$0 \text{ cm} \leq y \leq 24 \text{ cm}$$



$$h_0 = \sqrt{x_0^2 + y_0^2}$$

$$b = \arctan\left(\frac{y_0}{x_0}\right)$$



$$h_0 = \sqrt{x_0^2 + y_0^2}$$

$$\phi = \arctan\left(\frac{y_0}{x_0}\right)$$

Law of Cosines

$$a^2 = b^2 + c^2 - 2bc \cdot \cos(A)$$

$$b^2 = a^2 + c^2 - 2ac \cdot \cos(B)$$

$$c^2 = a^2 + b^2 - 2ab \cdot \cos(C)$$

$$\theta = \arccos\left(\frac{L_1^2 + L_2^2 - L_3^2}{2 \cdot L_1 \cdot L_2}\right)$$

$$\alpha_1 = \theta + \phi$$

$$L_1 = 120 \text{ mm}$$

$$L_2 = 120 \text{ mm}$$

$$\cos(90^\circ + \theta_2) = \frac{L_1^2 + L_2^2 - L_3^2}{2 \cdot L_1 \cdot L_2}$$

$$-\sin \theta_2 = \frac{L_1^2 + L_2^2 - L_3^2}{2 \cdot L_1 \cdot L_2}$$

$$\theta_2 = \arcsin\left(\frac{L_1^2 + L_2^2 - L_3^2}{2 \cdot L_1 \cdot L_2}\right)$$

$$\alpha_2 = 180^\circ - \theta_2$$



$$h = \sqrt{x_0^2 + y_0^2}$$

$$b = \arctan\left(\frac{y_0}{x_0}\right)$$



$$h = \sqrt{x_0^2 + y_0^2}$$

$$\phi = \arctan\left(\frac{y_0}{x_0}\right)$$



Law of Cosines

$$a^2 = b^2 + c^2 - 2bc \cdot \cos(A)$$

$$b^2 = a^2 + c^2 - 2ac \cdot \cos(B)$$

$$c^2 = a^2 + b^2 - 2ab \cdot \cos(C)$$

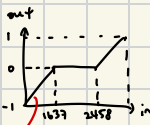
$$\theta_1 = \arccos\left(\frac{L_1^2 + L_2^2 - L_3^2}{2 \cdot L_1 \cdot L_2}\right)$$

$$\theta_2 = \arccos\left(\frac{L_1^2 + L_2^2 - L_3^2}{2 \cdot L_1 \cdot L_2}\right)$$

$$\alpha_1 = \theta_1 + \phi$$

$$\alpha_2 = 360^\circ - 135^\circ - \theta_2$$

$$0 \leftarrow \frac{2047}{2048} \rightarrow 4095$$



$$\text{out} = -1 + \frac{0 - (-1)}{4095 - 0} \cdot \text{in}$$

$$y = mx + c$$

$$0 = \frac{1 - 0}{4095 - 2458} \times 2458 + c$$

$$\text{out}_2 = \frac{1 - 0}{4095 - 2458} \times \text{in} + \left(-\frac{1 - 0}{4095 - 2458} \times 2458\right)$$

$$= \frac{1 - 0}{4095 - 2458} \times (\text{in} - 2458)$$

$$0 \sim 4095$$

$$0 \sim (2047 - \pi), (2047 + \pi) \sim 4095$$

$$m = 440$$

$$n = 205$$

$$[0, 1842] \cup [2252, 4095]$$



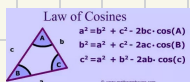
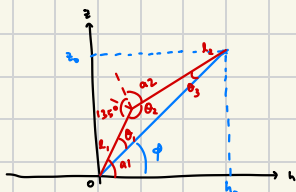
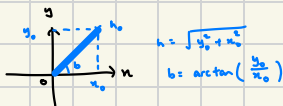
$$y = mx + c$$

$$0 = \left[\frac{100 - 0}{4095 - 2252}\right] \cdot 2252 + c$$

$$c = -122$$

$$f(x) = \begin{cases} -100 + \left[\frac{0 - (-100)}{1842 - 0}\right]x, & 0 \leq x \leq 1842 \\ 0, & 1842 < x \leq 2252 \\ \left[\frac{100 - 0}{4095 - 2252}\right] \cdot 2252 + \left[\frac{100 - 0}{4095 - 2252}\right]x, & 2252 \leq x \leq 4095 \end{cases}$$

undefined, otherwise



$$\theta_1 = \arccos\left(\frac{h_x^2 + h_z^2 - h^2}{2 \cdot h_x \cdot h_z}\right)$$

$$\theta_2 = \arccos\left(\frac{h_x^2 + h_y^2 - h^2}{2 \cdot h_x \cdot h_y}\right)$$

$$a1 = \theta_1 + \phi$$

$$a2 = 360^\circ - 135^\circ - \theta_2$$



$$\gamma = (90 - \phi) - (90 - \theta_2)$$

$$= -\phi + \theta_2$$

$$\gamma = \theta_2 - \phi$$

$$\gamma = (180^\circ - \theta_1 - \theta_2) - \phi$$

$$0 \leftarrow \frac{2047}{2048} \rightarrow 4095$$



$$\text{out} = -1 + \frac{0 - (-1)}{1637 - 0} \cdot \text{in}$$

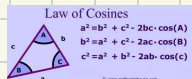
$$y = mx + c$$

$$0 = \frac{1 - 0}{4095 - 2458} \times 2458 + c$$

$$\text{out} = \frac{1 - 0}{4095 - 2458} \times \text{in} + \left(-\frac{1 - 0}{4095 - 2458} \times 2458\right)$$

$$= \frac{1 - 0}{4095 - 2458} \times (\text{in} - 2458)$$

input x - angle,
 x - length,
 y - length



$$h = \sqrt{x_0^2 + y_0^2}$$

$$\phi = \arctan\left(\frac{y_0}{x_0}\right)$$

$$\theta_1 = \arccos\left(\frac{l_1^2 + h^2 - l_2^2}{2 l_1 \cdot h}\right)$$

$$\theta_2 = \arccos\left(\frac{l_1^2 + l_2^2 - h^2}{2 l_1 \cdot l_2}\right)$$

$$\alpha_1 = \theta_1 - \phi$$

$$\alpha_2 = 360^\circ - 135^\circ - \theta_2$$

$$\frac{\text{pulse}}{20\text{ms}} = \frac{x}{1000}$$

$$\frac{\text{pulse}}{20000\mu\text{s}} = \frac{x}{1000}$$

$$x = \frac{\text{pulse}}{20000\mu\text{s}} \times 1000$$

$$0 \leftarrow \frac{2047}{2048} \rightarrow 4095$$



$$\text{out} = -1 + \frac{0 - (-1)}{2047 - 0} \times \text{in}$$

$$y = mx + c$$

$$0 = \frac{1 - 0}{4095 - 2458} \times 2458 + c$$

$$\text{out} = \frac{1 - 0}{4095 - 2458} \times \text{in} + \left(-\frac{1 - 0}{4095 - 2458} \times 2458 \right)$$

$$= \frac{1 - 0}{4095 - 2458} \times (\text{in} - 2458)$$

$$0 \rightarrow 2047, 2048 \rightarrow 4095$$

$$\text{With } 1\% \text{ deadtime: } 0 \rightarrow 2047 - \left(\frac{4095}{100} \times 2 \times \frac{1}{2} \right),$$

$$2048 + \left(\frac{4095}{100} \times 2 \times \frac{1}{2} \right) \rightarrow 4095$$

$$\text{Let } LB = 2047 - \left(\frac{4095}{100} \times 2 \times \frac{1}{2} \right)$$

$$UB = 2048 + \left(\frac{4095}{100} \times 2 \times \frac{1}{2} \right)$$



$$m = \frac{0 - (-1)}{UB - 0} = \frac{1 - 0}{4095 - LB}$$

$$n = \frac{1}{2047 - \left(\frac{4095}{100} \right) \times \frac{1}{2}}$$

if ($\text{in} < LB$)

$$\text{out} = -1 + (m)(n)$$

else if ($\text{in} > UB$)

$$\text{out} = -1 + (m)[\text{in} - (UB - LB)]$$

else

$$\text{out} = 0$$